Column is 4m. Assume yield stress in Steel is 2500 N/mm^2 (CO2)

Q.38 Explain various types of trusses with neat sketch. (CO7)

SECTION-E

Note: Long answer type questions. Attempt any two questions out of three questions. (25x2=50)

Q.39 Draw front and side elevation of splicing arrangement of two unequal columns having different flange width with the following data: (CO2)

Lower column=ISHB300 @ 63.0 kg/m.

- Upper column =ISHB250 @51kg/m.
- Distribution plate =300mmx200mmx25mm
- Packing plate thickness =25mm
- Cover plate = 200mmx600mmx20mm
- Nominal diameter of rivets = 18 mm

Q.40 Draw the elevation of a column head connection of a column and beam with the following data: (CO3)

Column= ISMB 250 @ 37.3Kg/m

Beam = ISLB 250 @ 27.9Kg/m

Distributing plate = 400 x 125 x 12mm

Cleat = ISA150X150X10mm Rivets diameter - 18mm

Q.41 Draw a detailed drawing of simple fink steel roof truss from the following data (CO2)

Effective span= 6.50m

Pitch =30^o

Principal rafter - ISA 65X65 mm

Tie= ISA 65X65 mm

Struts=ISA 50X50 mm

Wind tie= 40 x 6mm

Cleats = ISA100X65X8

Shoe angle ISA 75X75 -2 nos

Rag bolt = 15 mm dia . And 150 mm long

AC sheet roof covering. Any other data not given may be assumed.

No. of Printed Pages : 4 180761/030761/0762

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6th Sem / Civil, Brick Tech., Constr. Mgmt.
Subject:- Steel Structures Design and Drawing

Time : 6Hrs. M.M. : 150

SECTION-A

Note: Multiple choice questions. All questions are compulsory (10x1=10)

Q.1 Rolled steel beams are used to resist (CO1)

a) Bending stress b) Tensile stress

c) Compressive stress d) All of the above

Q.2 The most commonly used rivet head is (CO2)

a) Snap b) Pan

c) Flat d) None of the above

Q.3 For structure under fatigue stresses, the preferred joint is (CO3)

a) Riveted b) Welded

c) Both d) None of the above

Q.4 Net area of plates connected by chain riveting is given by the relation (CO4)

a) (b-d) x t b) (b-nt) x t

c) (b-nd) x t d) none of the above

Q.5 For design of steel structure following I.S-code is used (CO5)

a) 800-2005 b) 800-2007

c) 800-2004 d) None of these

Q.6 Unit of radius of gyration is (CO5)

a) mm b) mm^2

c) mm^3 d) mm^4

Q.7 In a roof trusses the roofing material is supported by (CO6)

a) Principal rafter b) Purlin

c) Bottom chord d) None of the above

Q.8 The load carrying capacity is more in (CO8)
a) Long column b) Short column
c) Medium column d) All of the above

Q.9 Maximum shear stress is equal to (CO9)
a) 0.45fy b) 0.35 fy
c) 0.65fy d) 0.55 fy

Q.10 Maximum deflection for a simply supported beam should not exceed by (CO9)
a) I/225 b) I/325
c) I/425 d) I/525

SECTION-B

Note: Objective type questions. All questions are compulsory. (10x1=10)

Q.11 Brittleness in steel is because of______ (CO1)

Q.12 Nominal Diameter of a rivet is also called ________ (CO2)

Q.13 Lap, tee or corner joints require______ weld (CO3)

Q.14 The strength of tension member depends on its ______ (CO4)

Q.15 Strut is a member in Tension(T/F) (CO5)

Q.16 Roof trusses are economical for span more than______ (CO6)

Q.17 ______ Column fail due to purely crushing. (CO7)

Q.18 Bending Stress at Neutral axis is_______ (CO8)

Q.19 The process of assembling the fabricated components on site is called_____ (CO9)

Q.20 Concrete bed is economical where ______ of foundation is considerably high. (Co10)

SECTION-C

Note: Short answer type questions. Attempt any twelve questions out of fifteen questions. (12x5=60)

Q.21 What are the different ways in which riveted joint may fail? (CO2)

Q.22 Explain five advantages of bolt connection over riveted joints. (CO3)

Q.23 What is Pitch, edge distance and gauge is bolt connection (CO4)

Q.24 Calculate the Strength of ISA 100 X 65 X 8mm which is used as tension member with its longer leg connected at its end by 16 m diameter riveted. (CO5)

Q.25 What are the various types of sections used as torsion member? (CO5)

Q.26 Enlist the steps involved in the design of axial compression member. (CO6)

Q.27 A strut in a roof truss carries an Axial Compressive load of 180KN. Design a suitable double angle section for a member. The length of strut between Centre to centre of intersection is 2.5. (CO6)

Q.28 Define radius of gyration and slenderness ratio. (CO6)

Q.29 What are the factors affecting selection of roof truss. (CO8)

Q.30 What do you mean by fabrication. (CO10)

Q.31 Write the steps for design of columns. (CO8)

Q.32 Enlist the steps used in calculating load carrying capacity of a beam (CO9)

Q.33 How struts are different from columns. (CO5)

Q.34 Enlist any five properties of Structural steel. (CO1)

Q.35 Differentiate between lap joint and butt joint. (CO2)

SECTION-D

Note: Long answer type questions. Attempt any two questions out of three questions. (2x10=20)

Q.36 Calculate the effective throat thickness of the following fillet welds. (CO2)
i) 8 m fillet welt
ii) A fillet weld of one leg 6 mm & other 8 mm.

Q.37 Calculate the load carrying capacity of ISMB 400@ 604 N/m to be used as a column. the effective length of